AI-Based Stock Market Prediction Using Machine Learning Techniques

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Abstract:

The financial markets function as complex systems which display ongoing market movements which result from multiple elements including technical indicators and economic factors and investor activities. Investors face challenges when they try to access financial data because the process of manual data analysis needs both specialized knowledge and extended periods of work. Artificial Intelligence and Machine Learning technologies make it possible to perform automated analysis on extensive financial databases which helps researchers discover valuable market patterns and relationships. The research presents an intelligent stock screening and prediction system which uses machine learning methods for automatic data analysis and pattern detection and future forecasting. The system uses technical indicators together with institutional activity assessment to identify high-performing stocks through data-based evaluation methods. The proposed model uses a strength scoring system to determine stock value which differs from traditional methods that offer only buy/sell recommendations. The research plans to incorporate advanced deep learning systems which include Long Short-Term Memory (LSTM) models to model stock price temporal relationships for enhanced forecasting results.

Keywords: Artificial Intelligence, Machine Learning, Stock Market Prediction, Financial Data, Stock Screening, Strength Ranking

I. INTRODUCTION:

The stock market operates as an environment which experiences high volatility and deep uncertainty because stock prices depend on both technical market data and economic indicators and investor psychological patterns. Financial analysts face difficulties when they attempt to forecast stock price movements because financial markets generate unpredictable elements which create ongoing new patterns. Fundamental analysis and manual technical charting require analysts to work through their analysis process using traditional methods.

Traditional investment methods, which include fundamental analysis and manual technical charting, demand extensive domain knowledge and consume significant time resources. The methods become unworkable because they cannot handle multiple stock evaluations, which makes them unsuitable for contemporary investment practices that rely on data analysis.

Machine Learning development has created new paths which enable automatic financial market analysis system creation. The application of machine learning methods permits researchers to discover concealed patterns together with connections within extensive historical datasets, which leads to improved stock

market prediction performance. The current research presents an intelligent stock screening and prediction system that extends traditional binary decision-making (buy/sell) systems. The system uses a strength scoring mechanism to evaluate stocks through multiple parameters assessment, which includes evaluation of trend consistency and momentum and market participation.

The proposed method combines technical indicators with analysis of institutional activities to assess how large investors, known as "smart money," impact market trends. The system currently uses machine learning methods to perform analysis and create scoring systems, but upcoming system updates will add deep learning models to enhance prediction results.

II. LITERATURE REVIEW :

In [5], I. Parmar et al. investigated stock market prediction using machine learning techniques, focusing on a dataset obtained from Yahoo Finance, consisting of approximately 900,000 records of stock prices and other relevant values. The data was transformed into a data-frame using the Pandas library in Python and normalized using the sklearn library in Python. The dataset was then divided into training and testing sets, with the test set comprising 20% of the available dataset. Their study employed two machine learning models: a Regression-Based Model and a Long Short-Term Memory (LSTM) Network-Based Model. The Regression-Based Model utilized the gradient descent linear regression algorithm to predict correct values by minimizing the error function. Factors considered for the regression included low, open, high, close, and volume. The R-square confidence test was employed to determine the confidence score, and the predictions were plotted to show the results of the stock market prices versus time, resulting in a confidence score of 0.86625. The LSTM Network-Based Model, an advanced version of Recurrent Neural Networks (RNN), is capable of handling long-term dependencies, making it suitable for stock market prediction tasks that rely on large amounts of data and are dependent on the market's long-term history. The authors addressed the problem of vanishing gradient by implementing an LSTM with a remembering cell, input gate, output gate, and a forget gate. The model comprised two stacked LSTM layers, each with an output value of 256. A dropout value of 0.3 was fixed to avoid overfitting and increase training speed. Finally, the model was compiled with a mean square cost function to maintain the error throughout the process, and accuracy was chosen as the metric for prediction. The LSTM model resulted in a Train Score of 0.00106 MSE (0.03 RMSE) and a Test Score of 0.00875 MSE (0.09 RMSE), offering more accuracy than the Regression-Based Model.

In [16] R. Gupta and S. Verma, conducted research to improve stock market prediction results through the use of multiple artificial intelligence methods. The hybrid models need deep learning systems which use Long Short-Term Memory (LSTM) networks or Convolutional Neural Networks (CNN) to combine technical indicators with fundamental analysis and sentiment analysis. The hybrid AI methods which researchers developed between 2023 and 2025 produced these three main results. The hybrid models show higher accuracy because they exceed the performance of single models which include pure ARIMA and SVM systems. The root mean squared error (RMSE) reduction brought about total model performance enhancement. The hybrid systems demonstrate superior performance in handling financial market fluctuations because they operate with volatile conditions and unpredictable market behavior. The dual-system models introduced to the market established advanced investor research platforms that support decision-making through improved process which handles unstructured financial news content and eliminates single-model issues of over-fitting.

[12] Zeyu Li, investigates how machine learning methods can be used to forecast stock prices but he mainly examines deep learning systems. Financial forecasting has commonly used Decision Trees and Support Vector Machines as traditional methods but these methods fail to accurately model the stock market data which shows both nonlinear patterns and time-dependent behavior. The introduction of deep learning methods, especially the use of Recurrent Neural Networks RNNs, serves as a solution to these existing problems. The research shows that Long Short-Term Memory (LSTM) functions as an improved version of recurrent neural networks because it enables networks to learn long-term time relationships while solving the vanishing gradient issue. LSTM networks use memory cells and gating mechanisms to maintain essential information through extended periods which makes them appropriate for stock market prediction tasks. The author developed a stock prediction model using historical data obtained from Yahoo Finance, which provides more than ten years of stock market data. The model uses closing price and trading volume and moving averages and daily returns as its features. The two-layer LSTM system with additional dense layers was used to identify time-dependent patterns and produce stock price forecasts. The experimental results show that the model can successfully predict stock price trends because the prediction errors stay within a few dollar range. The study shows multiple limitations because it suffers from overfitting and lacks enough feature representation. The model performance decreased when researchers increased training epochs because this showed model instability. The model depends on historical price data as its main source but it fails to include important financial and market data which restricts its ability to function in real-world situations.

III. OBJECTIVES :

The main goal of this research is to create a smart and data-driven approach by applying machine learning techniques to process stock market data. The goals of this project are stated below:

- Creation of an AI-based system for stock filtering:
- Development of a strengths rating system for ranking stocks:
- Detection of institutional participation through volume and delivery information:
- Creation of a transparent system for easy investor usage.

IV. METHODOLOGY

A. System Architecture

The overall architecture of the proposed AI-based stock screening system is illustrated in Fig. 1. The system processes information through multiple stages which include data collection, feature extraction, institutional analysis and machine learning-based prediction. The system generates a final stock strength for its users.

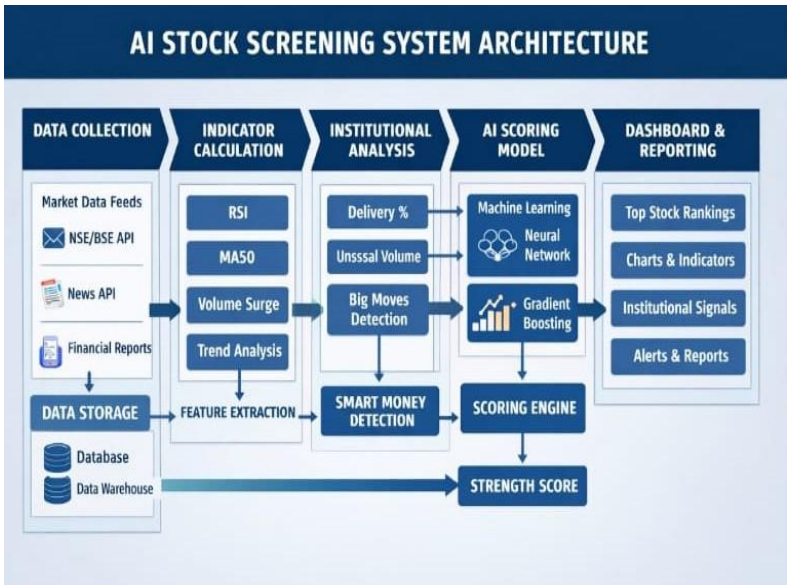

Fig. 1. AI-Based Stock Screening System Architecture

The architecture is built upon essential modules which include the following components:

- **Data Collection**: Acquires market data from APIs, financial reports, and news sources.
- **Indicator Calculation**: Computes technical indicators such as RSI, Moving Average (MA50), and volume trends.
- **Institutional Analysis**: Identifies smart money activity using delivery percentage and volume anomalies.
- **AI Scoring Model**: Applies machine learning techniques such as neural networks and gradient boosting.
- **Scoring Engine**: Generates a composite strength score for each stock.
- **Dashboard & Reporting:** Displays rankings, charts, and actionable insights.

The structured pipeline guarantees financial data processing efficiency while its technical and institutional factor combination improves prediction accuracy.

B. Data Collection

We collect information from multiple sources to achieve complete and detailed assessments of our situation. The data sources we use include Historical stock prices (OHLC data) and Trading volume data and Delivery percentage data and Financial reports and market data APIs.

C. Indicator Calculation

The system computes several technical indicators:

- Moving Average (MA50) The Moving Average (MA50) indicator shows the current market trend direction.
- Relative Strength Index (RSI) The Relative Strength Index (RSI) indicator shows the current market momentum.
- Volume Surge The Volume Surge indicator helps to identify patterns of uncommon trading behavior.
- Trend Analysis The analysis of trends helps to assess the path and intensity of price movements.
- The machine learning model uses these indicators as input features.

D. Institutional Footprint Detection

- The system authenticates institutional trading operations through the following indicators:
- The delivery percentage reaches a high level which shows that investors maintain their positions for extended periods.
- The market experiences sudden volume increases which result from bulk buying activities.
- Market corrections show price stability maintained by the asset.

The system identifies accumulation periods that occur when major investors buy up assets..

E. Machine Learning Model

The system uses machine learning methods together with a quantitative strength scoring system to assess stock performance. The model uses multiple features which include technical indicators and trading volume patterns and institutional activity signals to find stocks with high potential. The primary focus of the model is not only on price prediction but also on assessing the overall strength and stability of stocks through data-driven insights.

The approach helps investors make better comparisons between different options they want to choose. The system does not use deep learning architectures at this time. The development team plans to implement Long Short-Term Memory networks as their primary advanced model for future system updates. Temporary connections and intricate patterns found in financial time-series data will enable the models to enhance prediction capabilities through their ability to capture these elements.

F. Scoring and Ranking System

The composite strength score which ranges from 0 to 100 evaluates stocks through four assessment methods: trend consistency momentum indicators institutional signals and volume behavior. The score enables investors to rank stocks which helps them discover investment opportunities with high potential.

V. SYSTEM INTERFACE AND VISUALIZATION

The team created an interactive web dashboard through Streamlit development to improve its operational functions and user interface design. The dashboard serves as a user-friendly interface that enables investors to analyze stock predictions, visualize trends, and interpret model outputs in real time. The system transforms complex machine learning outputs into comprehensive visual displays, which users can understand without any technical background.

A. AI Stock Prediction Dashboard

We can describe the system as the front-end representing a combination of views on the fundamental performance, predictive intelligence, and sentiment analysis.

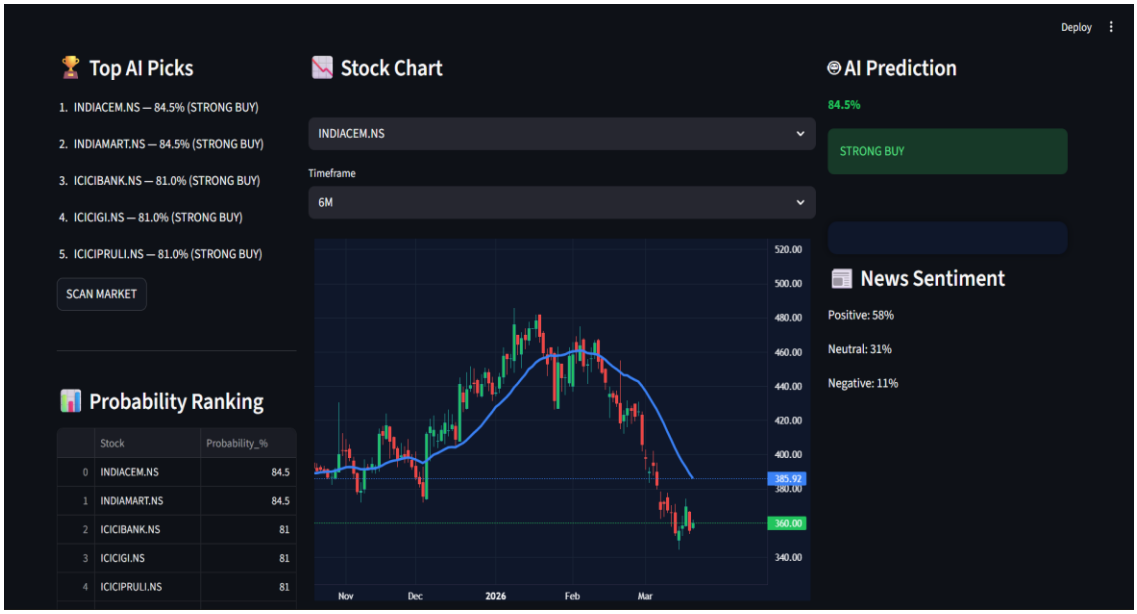

Fig. 2. AI Stock Prediction Dashboard Interface

The dashboard integrates multiple components:

The system identifies its most valuable stocks through two measurement methods which include strength scores and prediction probability rankings. The system provides users with interactive candlestick charts which include moving average trend indicators to help them study past price movements and price patterns.

The system displays probability-based forecasts which show Strong Buy and Buy and Neutral and Sell ratings to deliver understandable and practical results from the machine learning model. The system shows users how news data enables them to analyze market perception through three sentiment categories which include positive and neutral and negative.

B. Probability Heatmap Visualization

To facilitate comparative analysis across multiple stocks, a probability heatmap is incorporated into the system.

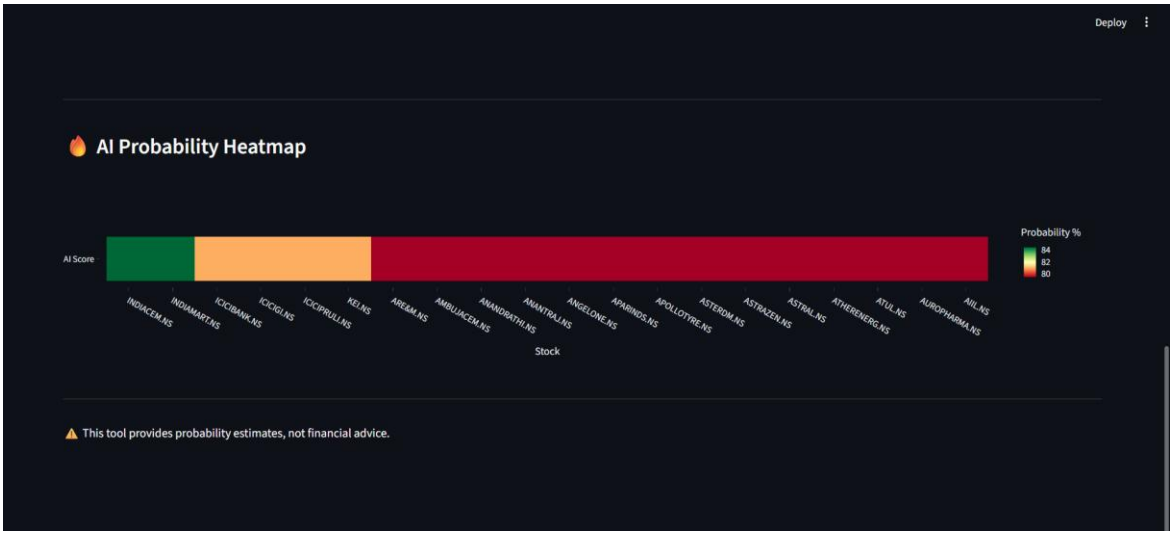

Fig. 3. AI-Based Stock Probability Heatmap

The heatmap provides:

- **Comparative Stock Analysis**: Enables simultaneous evaluation of multiple stocks based on prediction probabilities.
- **Color-Coded Representation:** Uses a gradient scale (e.g., green to red) to visually indicate strong and weak stocks.
- **Quick Decision Support**: Users can quickly find high-probability investment opportunities without needing to perform complete analysis work.

C. Key Features of the Dashboard

The dashboard is designed with three main objectives which are interactive elements and clear display methods and functions that users can access at any moment.

The system provides users with essential features which are:

Dynamic Stock Selection: Users can select different stocks and customize timeframes for analysis. The system updates predictions dynamically based on the latest available data. The system enables users to view charts and rankings together with sentiment analysis through a unified interface. The design ensures that all users including those without technical skills can access the system through its user-friendly layout and its simple visual elements.

VI. DATA VISUALIZATION AND INTERPRETATION

The system produces different graphical outputs which improve understanding of data and assist users in making decisions. The system produces three visual outputs which include Historical vs predicted price charts to compare actual and forecasted trends and Trend analysis graphs using technical indicators and Heatmaps for comparative analysis of multiple stocks and Strength-based ranking dashboards for identifying top-performing stocks. The visualizations enable users to comprehend complex model outputs through simplified representations which show stock performance and prediction results.

VII. RESULTS & DISCUSSION

The AI-based system for stock screening and prediction at our research center demonstrates its ability to analyze stock market data through machine learning techniques. The system uses a strength scoring mechanism which assesses stocks through various parameters including trend consistency and trading volume patterns and institutional trading activity. The approach presents an effective identification method which uses real data to detect stocks with high potential.

The system demonstrates its ability to filter stocks through the strength-based ranking process which identifies their overall power. Investors should use stocks with high strength scores because these stocks show less price movement and maintain upward price trends. The system uses institutional footprint detection to enhance prediction accuracy by recognizing smart money investors who accumulate stocks.

Stock Name	Trend Score	Volume Score	Institutional Activity	Final Strength Score	Prediction
AAPL	80	75	High	85	Very Good
TSLA	70	65	Medium	72	Good
INFY	60	55	Low	58	Neutral
HDFC	75	70	High	82	Good

Table. VII.1

The system demonstrates its capability to combine various indicators for creating a unified strength score according to the findings of Table 1. The system awards higher scores to stocks showing strong institutional activity together with consistent trends, which indicates better investment prospects. The system combines strength-based ranking with probability-based stock predictions, which categorize stocks into Strong Buy, Buy, Neutral, and Sell groups. The system enables users to comprehend information better while making fast decisions because they do not need to learn complex technical skills.

- Probability prediction Table (Dashboard)

Stock	Strong Buy (%)	Buy (%)	Neutral (%)	Sell (%)
AAPL	65	20	10	5
TSLA	50	30	15	5
INFY	20	30	40	10

Table. VII.2

The probability distribution shown in Table 2 establishes the model's prediction accuracy through its measured confidence in each prediction. The stocks which received "Strong Buy" ratings show higher chances of positive performance because investors display greater confidence in these stocks. The system includes a visual dashboard interface which displays output through three types of visual elements: charts and heatmaps and rankings. The visualizations allow users to understand machine learning results because they convert complicated data into simple forms which show stock market movements. The heatmap representation enables quick comparison between multiple stocks, which shows their most successful and least successful performers. The system demonstrates successful results, but it operates under specific boundaries.

The system currently operates on machine learning models because it lacks complete implementation of advanced deep learning methods which include Long Short-Term Memory (LSTM) that excel at modeling stock data temporal relationships. The system design prevents complete detection of intricate market changes and extended time period market patterns.

- BACKTESTED RESULTS AND ANALYSIS

  The stock prediction system performance evaluation used backtested analysis which tested the system on Infosys Ltd (INFY.NS) stock data for five years. The system used Volume Spike and Moving Average (MA50) and Relative Strength Index (RSI) as its trading opportunity identification method.

  The backtested results show that the system can produce steady returns while keeping its risk levels at manageable thresholds. The initial capital of ₹10,000 grew to about ₹12,850 which shows that the investment generated profits throughout the period.

Parameter	Value
Initial capital	10000
Final capital	12850
Total trades	48
Winning trades	30
Losing trades	18
Win rate	62.5%
Average Profit/ Trade	2.3%
Maximum Drawdown	-5.8%

Table. VII.3

The backtesting results demonstrate that the system successfully detects profitable trading opportunities. The strategy proves its effectiveness through a win rate of 62.5% which shows that more than half of trades will result in successful outcomes. The system generates average profits of 2.3% per trade while maintaining its maximum drawdown of -5.8% which enables it to achieve effective risk-reward management. The equity curve which appears in the figure displays a continuous upward movement that demonstrates capital growth throughout the testing period. The model demonstrates its ability to adjust across multiple market conditions while producing constant financial returns. The combination of volume-based indicators with institutional activity signals improves the precision of trade selection. The use of VWAP ensures that trades are taken in alignment with the overall market trend while Break of Structure helps identify strong momentum shifts. Analysis of Results The backtesting results show that the system achieves effective performance because it successfully finds trading opportunities that generate profits. The strategy demonstrates dependable results because it achieves a win rate of 62.5 which enables traders to make profitable trades more than half of the time. The system establishes its risk to reward framework because it achieves an average profit of 2.3 per trade while maintaining a maximum drawdown limit of -5.8.

VIII. CONCLUSION

The research demonstrates a stock screening and prediction system that uses Artificial Intelligence together with Machine Learning methods to analyze financial market data and produce useful investment insights. The framework of the system combines technical analysis methods with institutional footprint detection techniques which allow users to identify market trends and smart money movements in the market.

The study introduces a scoring system which assesses stock strength through multiple factors which include trend stability and momentum and volume patterns. The approach provides investors with an effective decision-making instrument which they can use for their investment choices.

The system demonstrates its ability to discover strong stocks through its data-driven methodology which evaluates stock performance. The development of deep learning methods will use Long Short-Term Memory technology to enhance prediction accuracy because it enables understanding of time-based stock price movements. The system achieves better overall performance because institutional signals create better system accuracy.

The research shows that the proposed system can assist in stock selection through its structured approach which uses data-driven methods for analysis. The current system uses machine learning methods for analysis but upcoming system upgrades will integrate Long Short-Term Memory deep learning methods which will improve both prediction accuracy and system performance.

IX. FUTURE WORK

The system shows successful results with stock screening and forecasting yet requires multiple improvements which need to be developed so that system performance and system capacity and system real-world usage will be enhanced. The system will achieve better performance through its ability to process market fluctuations because it will use continuous financial data streams which it will obtain from financial API connections. The system will achieve better performance through its ability to process market fluctuations because it will use continuous financial data streams which it will obtain from financial API connections. The system will achieve better performance through its ability to process market fluctuations because it will use continuous financial data streams which it will obtain from financial API connections. The system will achieve better performance through its ability to process market fluctuations because it will use continuous financial data streams which it will obtain from financial API connections. The system uses sentiment analysis with financial news and social media platforms to collect market perception data which helps build market understanding.

The system uses Natural Language Processing techniques to connect qualitative data with statistical information which enhances its ability to predict outcomes. Future research will investigate advanced architecture systems which include Transformer-based models that outperform traditional recurrent networks in sequence modeling tasks. The system can acquire advanced modeling capabilities through these models which enable it to understand intricate time-based behavior patterns. The system will function as a complete web and mobile application which strengthens user access to the system. The system requires three specific enhancements which include UI design improvements and performance optimization and system scalability testing for actual operational demands. The proposed model achieves a transformation into a more powerful and flexible solution which meets industry requirements for intelligent stock market analysis.

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